

pH Control

Part 1: Peihan Liu

Part 2: Gretna Truong

Part 3: Brian Chu

Part 4: Nicole Gorostiza

Section B01 (M/W), Team 11

Abstract

pH control is an essential process in wastewater treatment and bioreactor systems. The optimization of controller modes in PID control and controller tuning are necessary to ensure an appropriate the closed loop response for the process. Controller tuning based on well-established tuning methods using parameters from simple mathematical models, namely first order plus dead time is preferred. A closed system allowing the accumulation of acid and base in order to adjust pH in the presence of a buffer was made to determine the optimal controller mode and parameters. This process was found to be self-integrating, and the best controller mode was found to be proportional-derivative control. In the closed loop response to basification from 2 pH to 7 pH, the optimal setting for a PD controller was found to be $K_C = 28 \pm 3$ and $\tau_D = 4.6 \pm 0.3$ seconds, resulting in a rise time of 15 seconds and a rise time of 30 seconds with an overshoot less than 0.1 pH.

1 Introduction

The call for greater economic efficiency and sustainability has prompted improvements in bioprocesses controls, especially in the biopharmaceutical and nutraceutical industries.¹ The type of process controls best for a certain process depends on factors including the type of bioreactor used and the catalyst. pH control systems are also essential in wastewater treatment. Wastewater from metal plating facilities, for instance, contain many heavy metals, oil, grease, and suspended solids at levels hazardous to the environment.² Heavy metals are removed through precipitation which requires multiple steps in which the pH must be adjusted.² Many continuous chemical and biological processes require the environment to be set at a certain pH. Unexpected behavior in the process parameter and signal noise can confound the measurement of pH and disturbances in the pH.³ In bioreactor systems, disturbances cause parameters including feed flow rate, temperature, pressure, agitation, and pH, and dissolved oxygen concentration to shift from the set point.¹ A buffer is used to resist great changes in ion concentration in the addition of acid and base to create a region where the change in pH per mass of acid or base added is linear. In controls for continuous systems, usually pH is considered the control variable which is manipulated by the base and acid flow rates while buffer flow rate is treated as a disturbance variable.³ Outside of the buffer certain range, the behavior of the pH process is nonlinear. Complicated models of pH processes have been proposed since 1972.⁴ In the years following this proposal, numerous nonlinear control strategies have been proposed based on complex process models as well as assumptions such as fictitious dissociation constants for weak acids.⁵ There are challenges to the implementation of these controllers. Later, such models were reduced to a first-order state equation.⁶ Simpler process models can be modified easily to fit extensions from the buffer region in the titration curve and adjust to unexpected changes such as the buffer flow rate changing or some other species being introduced disturbing the pH.

2 Background

One of the challenges faced in pH process control is the complexity of its model, typically due to non-linearity and time-variance in biochemical processes. For example, proportional integral derivative (PID) controllers are widely utilized in the process control at the industrial level due to their effectiveness. This effectiveness is a result of the numerous different methods that can be utilized for controller tuning, such as direct synthesis or the internal model control (IMC) method.⁷ Each tuning method typically has its own set of disadvantages that could cause set-point changes or load disturbances to result in extreme instability. Regardless of tuning accuracy, large response-to-settling time and oscillatory responses are common for processes outside the expected bounds of simplicity.⁸ In an effort to preserve the ease of PID-controllers while minimizing overshoots and oscillations, researchers have shifted interest towards designing and tuning pseudo-PID-controlled systems, especially in relation to auto-tuning. This concept as an industrial application would allow for an increase in profit as system response time is decreased. Systems with an integrator model and PD-controller have shown promising dynamic loop performance.⁹

Another pH controller optimization option involves PID-controllers in a double-control scheme with IMC, which would allow for an accurately modeled process across the full range of the pH scale. The use of one controller for set-point tracking and a controller for disturbance rejection would result in a more closely monitored system in the case of simultaneous changes in set-point and load disturbance.³

3 Theory

Process controllers are characterized and designed based on transfer functions. By using an open loop system, a relationship between the input actuator signal and the outlet system re-

sponse can be established as a function of time,

$$G_p = \frac{A(s)}{Y(s)} \quad (1)$$

where G_p is the process transfer function. This resulting open loop transfer function of this system displays a ramp-like response and suggests a First Order + Integrator (FOPI) model. The integrator model accounts for a semi-batch process, which is due to experimental design.

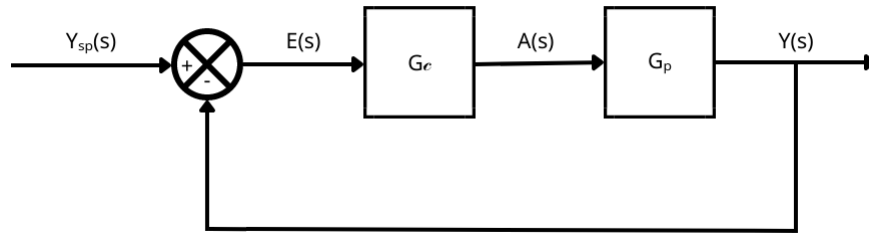


Figure 1: Assumed simplified block diagram of closed loop feed-back system where The proposed transfer function, is the set point pH within the buffer range, $A(s)$ is the percentage of power sent to each pump (actuator signal), $Y(s)$ is the observed response, and G_p is the transfer function.

If the process model is known or estimated, direct synthesis can be used to select a controller type as well as controller settings. Given that this system has a given integrator, controller type can be narrowed down to a PD-controller. A PD-controller's transfer function is

$$G_c = \frac{1}{K_p \tau_c} (1 + \tau_p(s)) \quad (2)$$

where K_p is process gain, τ_c is the time constant of the closed loop system, and τ_p is the process time constant. In this case, τ_p is also equal to τ_D , which is defined as derivative time. This controller type, PD, in combination with the observed process model, FOPI, resembles a psudeo-PID-controller type. Using this assumption, the process can be modeled as a first

order plus dead time model (FOPDT),

$$G_p = \frac{K e^{-\theta s}}{\tau s + 1} \quad (3)$$

where θ is process dead-time, $\mathbf{K}$ is process gain, and τ is the process time constant. This model can be used to fit the original data for $\mathbf{G}_p$ to solve for necessary PID coefficients. The assumption of a FOPDT model also allows for the implementation of Ziegler-Nichols (ZN) and Cohen-Coon (CC) tuning rules. Usage of these tuning rules allows calculation of coefficients in the equation of $\mathbf{G}_c$ for each controller type. The controller transfer function based on the FOPDT model is

$$G_c = K_c \quad (4)$$

when assuming a P-only controller,

$$G_c = K_c \left[1 + \frac{1}{\tau_I s} \right] \quad (5)$$

when assuming a PI-controller,

$$G_c = K'_c \left[1 + \frac{1}{\tau_I s} \right] \left[\frac{\tau_D' s + 1}{\alpha \tau_D' s + 1} \right] \quad (6)$$

when assuming a PID-controller, where proportional gain, $\mathbf{K}_c$, integral time, τ_I , and derivative time, τ_D . It is typical for number of oscillations to decrease as τ_I increases or as $\mathbf{K}_c$ decreases. It is also expected that an increase in $\mathbf{K}_c$ decreased initial overshoot.

4 Methods

The experiment was conducted by sending strong base and strong acid solutions through plastic tubes into a beaker with the a buffer via a peristaltic pump. Two peristaltic pumps were used to deliver 0.1M sodium hydroxide and 0.1M hydrochloric acid each into a stirred beaker

containing a solution containing phosphate buffered saline in excess, ensuring a titration curve buffer region from pH 5 to 7. Flow rates of up to 1.4 mLs^{-1} were used in adjusting the pH. pH was continuously monitored using an pH probe. The flow rate of acid and base were used to control the pH via set point changes, and the response over time was recorded in LabVIEW. Voltage change input was sent through LabVIEW to the pumps in order to adjust the flow rates. An open loop test of the process alone was performed. The response to a set point change of 6 pH to 7 pH was used to find first order plus dead time (FOPDT) parameters to use in controller tuning. FOPDT parameters Figs. 12 and 13 were used to tune a PID controller according to direct synthesis and Ziegler-Nichols tuning methods in a closed loop response. Responses to positive set point changes to 7 pH were analyzed to determine the coefficients for the best response of the PID controller based on overshoot, oscillation, and time to reach steady state. PID coefficients were also changed based on heuristics determined from the experiment.

5 Results and Discussion

The efficiency of controller settings may be determined upon minimal overshoot, oscillations, and rise and settle times depending on the machinery and its uses. In this case, pH control prioritizes overshoot and rise time to control wastewater systems. For consistency, the pH was changed from a start value of 2 to a setpoint value of 7. Other cases in the buffer region from above and below various setpoints were tested for and found to have similarity in system response.

5.1 FOPDT Tuning Correlations

The FOPDT parameters were determined by analyzing multiple ramp responses with a result of $K_P = 0.03 \pm 0.02 \frac{\Delta pH}{\text{Controller Output out of 100}}$, $\tau_P = 4.6 \pm 0.3 \text{ s}$, $\theta_P = 4 \pm 3 \text{ s}$. The large time delay, θ_P , can be due to siphoning issues since the pump will be off for some time and suddenly increases, pushing liquid particles where gas was and is limited by the pipe diameter. In addition this

delay may be from fouling in the pipes feeding in acid and base, similar to siphoning issues but for solid buildup, or lag input in the computer to pump.

The tuning methods for the controller were calculated using a FOPDT with Pade Approximation using Dahlin Synthesis and calculated for a PID and PD controller. It was also calculated using the IAE and Ziegler Nichols (ZN) tuning parameters. These tuning parameters and methods helped to find the optimal controller parameters. The rise time in this case is how long the system takes to reach setpoint value and the settle time is duration to reach setpoint without oscillations or overshoot.

The effects of variation of each parameter of a PD controller is analyzed to find optimal parameters using tuning parameters as start values.

5.2 Proportional Control

The tuning parameters were used to first determine a good controller gain using a P-Only controller.

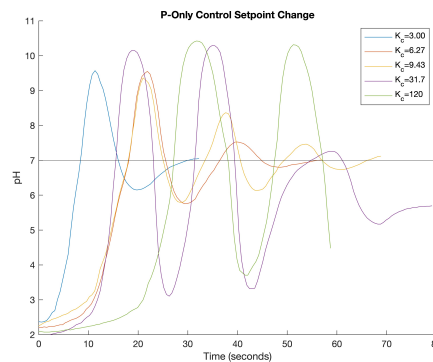


Figure 2: Comparison of Set-point Responses with Various Proportional Control Gain Values

Fig. 2 shows a various range of controller gains, K_C , where a lower gain shows less oscillations and a quicker rise and settle time. It is visible that the larger gains even have sustained oscillations as large as ± 3 pH, which is not optimal to stabilize pH to a certain value. It is clear that gain values of 3 (blue) and 6.27 (orange) are superior in settle time, with 3 having a shorter

rise time.

5.3 Various Controller Forms

5.3.1 Effects of Integral Control

Adding an integral controller is the next step, to remove steady state error as an accurate pH value is necessary for wastewater.

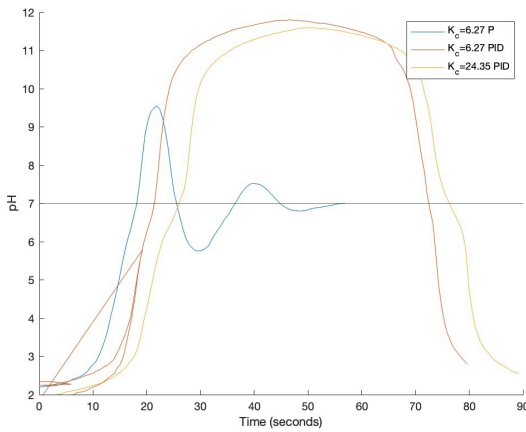


Figure 3: Comparing closed loop response with different controller settings while adding an integral control.

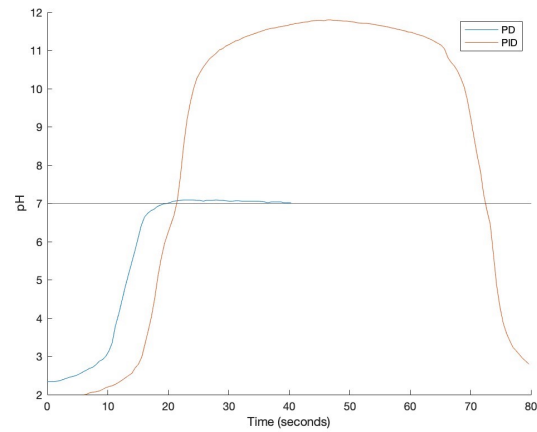


Figure 4: Comparing closed loop response using PD and PID controller with $K_c = 6.27$, $\tau_D = 0.05$ s, where PID adds an integral control of $\tau_I = 0.9$ s

It is seen above in Fig. 3 that adding an integral controller along with a derivative control creates a large overshoot and a large increase in rise and settle time and never returns to the setpoint. This was tested again using a PD and PID controller to isolate the result of adding an integral controller Fig. 4, where the same overarching shape appears. This is a result of this process being a self-integrating system as the pH will increase faster at a larger rate based on base added, rather than increasing in value. This means the integrator solves the issue of setpoint steady state error, therefore not needing an I-control in the controller for this process.

5.3.2 Variability of Derivative Control

The derivative time constant, τ_D gives us a variation in oscillation control and rise and settle time. Since larger controller gains have greater oscillations, the derivative control was increased to possibly withstand these oscillations in Fig. 9. It is seen that the oscillations are decreased from about 3 pH to 0.5 pH while yielding a similar rise time, but still sustain oscillations. This is due to the controller gain being too powerful, so a smaller K_C is optimal for our controller.

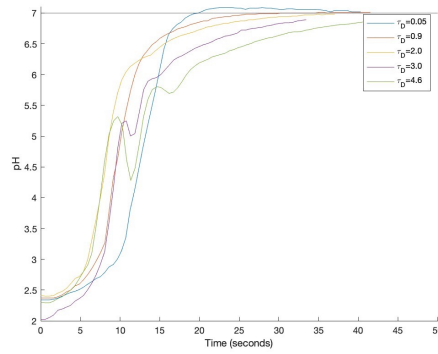


Figure 5: Variation of Derivative time setting on a PD controller using $K_C = 6.27$.

Fig. 5 shows that decreasing the derivative time allows a quicker rise time as the blue line using $\tau_D = 0.05$ seconds has a rise time of about 17 seconds. In addition, some of the larger values shows oscillations in the beginning of the system response with all values having similar settle times. This shows a smaller derivative time is optimal.

The close up Fig. 10 shows that too small of a τ_D can lead to slightly more overshoot and longer settle time with an upside of shorter rise time. Giving us an optimal value of $\tau_D = 0.05s$ for a controller gain of 6.27.

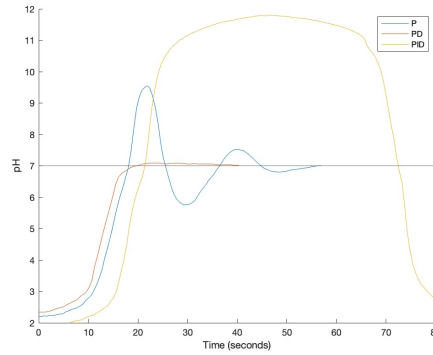


Figure 6: Setpoint Responses using Proportional-Only, Proportional-Derivative, and Proportional-Integral-Derivative Control

Fig. 6 analyzes P, PD, PID controllers. PI is not analyzed as an integrator system will see no advantage in adding an integrator which can mess with the self-integrating response. The PID is shown in comparison to clarify that adding I creates the shape leading to large overshoot. PD control is superior to P-Only control as it has nearly no oscillations and overshoot in comparison, a third of the settle time, and a similar rise time.

The smaller controller gain values were seen to have a smaller rise time, but yield too large of an overshoot with P-only control. Adding on derivative control, we see that the rise and settle time are increased greatly, meaning the controller gain of 3 is too weak in Fig. 11 to raise pH to setpoint with a PD controller.

5.4 Theoretical and Experimental Controller Settings

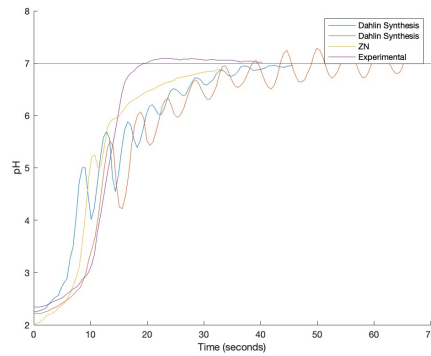


Figure 7: Comparison of Dahlin Synthesis and Ziegler-Nichols to Best Controller Settings

The time constant for the controller, τ_C , was chosen based on the PI and PID controller, but the best case here is a PD controller, so all tuning parameter correlations are tested against the best controller setting. The experimental controller settings is far superior in minimization of oscillations compared to Dahlin Synthesis, as well as rise and settle time [Fig. 7](#). The experimental settings have a rise time of about 19 seconds, while the Dahlin Synthesis yields a rise time of about 35 seconds and ZN with a rise time of 30 seconds. The settle time is faster in experimental data of about 37 seconds, with Dahlin Synthesis having about 45 seconds and ZN having 40 seconds.

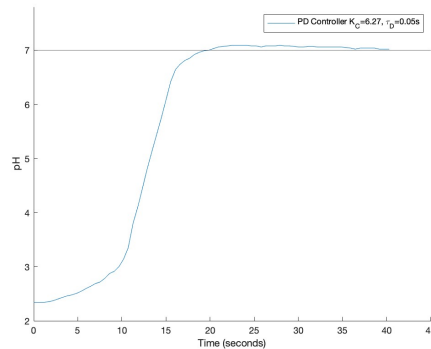


Figure 8: Optimal Controller Settings for pH Control

This controller setting yields a rise time of 15 seconds, settle time of 30 seconds, less than

0.1 pH overshoot, and no oscillations from Fig. 8 using a PI controller, $K_C = 6.27$, $\tau_D = 0.05$ seconds. The theoretical tuning parameters yield values of around $K_C = 28 \pm 3$ and $\tau_D = 4.6 \pm 0.3$ seconds, where the difference of the experimental gain and derivative time being around 25% and 10% of the theoretical values, respectively. This is due to the variability in measuring process parameters using FOPDT rather than modeling an integrator system. The FOPDT process parameters had more than 5% variability, deeming the values not significant, due to the variation of parameters depending on the pH range within the buffer region and jump in ramp response. Different jumps can lead to quicker or slower process time and time delays.

6 Conclusions

The optimal response for set point changes was found to come from the PD controller due to the rise time being shorter, the overshoot being smaller, and oscillations having smaller amplitude in cases where the integral time constant was introduced. The optimal steady-state gain and derivative time constant for the PD controller were determined to be $K_C = 28 \pm 3$ and $\tau_D = 4.6 \pm 0.3$ seconds. Whereas PD control was optimal, P-only control was found to have large oscillations and high settling time, and introducing integral control also caused large overshoots and increased the time scale to impractical values. These determinations were found to be due to the behavior of the buffer process being self-integrating. Further experiments should use controller tuning software to more accurately model the process and controller response and accordingly change PID parameters. Process dynamics should also be changed to examine pH change due to accumulation in the presence of an outlet flow rate. This experiment only tested basification starting from a pH of 2. Process dynamics and appropriate controller response for buffer behavior starting from a high pH reacting to acidification should also be examined. Furthermore, disturbance rejection response should also be tested in addition to set point changes.

Bibliography

- (1) Mitra, S.; Murthy, G. S. Bioreactor control systems in the biopharmaceutical industry: a critical perspective. *Systems Microbiology and Biomanufacturing* **2022**, 91–112.
- (2) Alwan, G. M. pH-Control Problems of Wastewater Treatment Plants. *Al-Khwarizmi Engineering Journal* **2008**, 37–45.
- (3) Shobana, S.; Panda, R. Control of pH process using double-control scheme. *Nonlinear Dynamics* **2011**, 2267–2277.
- (4) McAvoy, T.; Hsu, E.; Lowenthal, S. Control of pH in Controlled Stirred Tank Reactor. *Ind. Eng. Chem. Process Des. Dev.* **1972**, 11, 68–70.
- (5) Sung, S.; Lee, I. pH Control Using a Simple Set Point Change. *Ind. Eng. Chem. Res.* **1995**, 34, 1730–1734.
- (6) Wright, R.; Soroush, M.; Kravaris, C. Strong Acid Equivalent Control of pH Processes: An Experimental Study. *Ind. Eng. Chem. Res.* **1991**, 30, 2437–2444.
- (7) Anil, C.; Sree, R. Tuning of PID controllers for integrating systems using direct synthesis method. *ISA Transactions* **2015**, 57, 211–219.
- (8) Dey, C.; Mudi, R.; Simhachalam, D. A simple nonlinear PD controller for integrating processes. *ISA Transactions* **2014**, 53, 162–172.
- (9) Silveira, A. et al. Pseudo-PID Controller: Design, Tuning and Applications. *IFAC Proceedings Volumes (IFAC-PapersOnline)* **2012**, 2, 542–547.

Appendix

1. Bioreactor control systems in the biopharmaceutical industry: a critical perspective

- Author(s): S.Mitra, and G. Murthy
- Year published: 2022
- Journal name: Systems Microbiology and Biomanufacturing

(a) Review of industrial-scale commercial control systems strategies with a focus in biopharmaceutics. Discusses pH sensors and other technology to sense ions in solution.

2. pH-Control Problems of Wastewater Treatment Plants

- Author(s): G. Alwan
- Year published: 2008
- Journal name: Al-Khwarizmi Engineering Journal

(a) Found that PID system could treat most disturbances under normal operation in wastewater treatment plants

3. Control of pH process using double-control scheme

- Author(s): S. Shobana and R. Panda
- Year published: 2012
- Journal name: Nonlinear Dynamics

(a) Found an IMC-based PID double-controller scheme appropriate for set point both tracking and disturbance rejection in pH processes.

4. Tuning of PID controllers for integrating systems using direct synthesis method

- Author(s): C. Anil R.P. Sree

- Year published: 2015
- Journal name: ISA Transactions
- (a) Direct synthesis is used to design PID-controller parameters for integrating system and is found to accurately model a jacketed CSTR.
- (b) Proposed design was effective in modeling performance and input usage.

5. Pseudo-PID Controller: Design, Tuning and Applications

- Author(s): A.S. Silveira and A.R. Coelho and A.A. Franca and V.L. Knihs
- Year published: 2012
- Journal name: IFAC Proceedings Volumes
- (a) Proposed Integrator + PD controller design with ZN tuning parameters and IMC models a Pseudo-PID controller system and is determined to be relatively high performing when used in most digital technology.

6. A simple nonlinear PD controller for integrating processes

- Author(s): C. Dey and R.K. Mudi and D. Simhachalam
- Year published: 2014
- Journal name: ISA Transactions
- (a) Proposed auto-tuning proportional derivative controller (ADP) is more efficient than typical PD and PID controllers in conventional control loops as well as integrating processes.
- (b) ADP control is observed to be less efficient in closed-loop systems.

Trends of PD Controller Settings

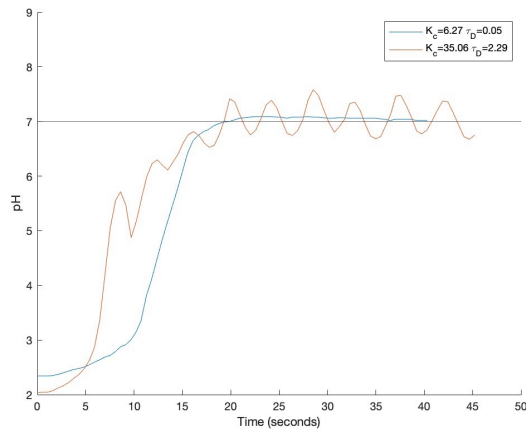


Figure 9: PD Controller with higher K_C and higher τ_D setting to account for oscillations

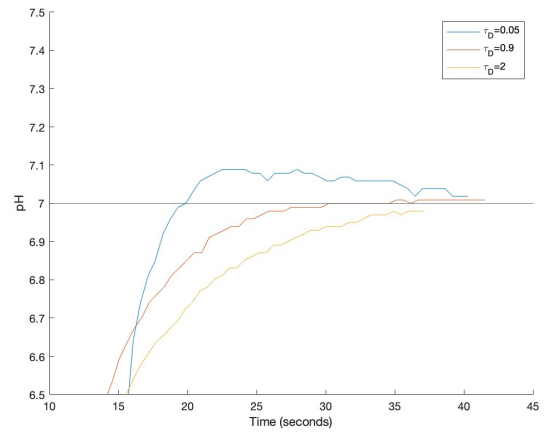


Figure 10: Zoomed in view of oscillations and overshoot of the optimal PD controller settings of $K_C = 6.27$

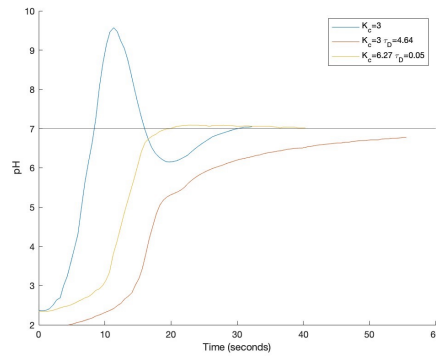


Figure 11: Comparison of smaller proportional gain values with a PD controller

Calibration Curves

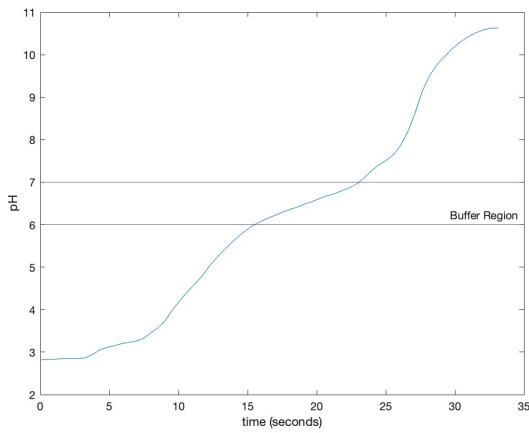


Figure 12: Titration Curve to Determine Buffer Region

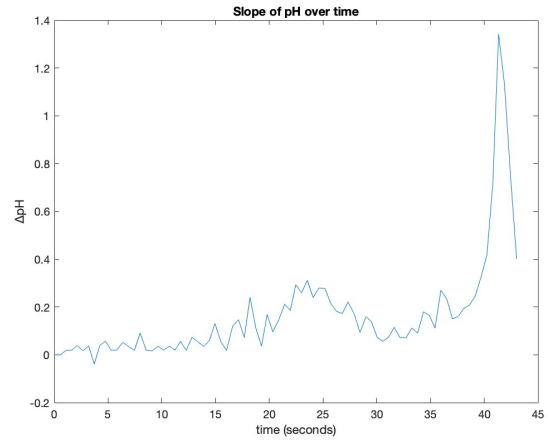


Figure 13: FOPDT Analysis on Ramp response

6.1 FOPDT Tuning Equations

Controller Type:	Proportional Gain:	Integral Time:	Derivative Time:
P-Only	$K_c = \frac{1}{K} \left(\frac{\tau}{\theta} \right) \left(1 + \frac{\theta}{3\tau} \right)$		
PI-Control	$K_c = \frac{1}{K} \left(\frac{\tau}{\theta} \right) \left(0.9 + \frac{\theta}{12\tau} \right)$	$\tau_i = \frac{\theta(30 + 3\theta/\tau)}{(9 + 20\theta/\tau)}$	
PD-Control	$K_c = \frac{1}{K} \left(\frac{\tau}{\theta} \right) \left(\frac{5}{4} + \frac{\theta}{6\tau} \right)$		$\tau_d = \frac{\theta(6 - 2\theta/\tau)}{(22 + 3\theta/\tau)}$
PID (actual)	$K_c' = \frac{1}{K} \left(\frac{\tau}{\theta} \right) \left(\frac{4}{3} + \frac{\theta}{4\tau} \right)$	$\tau_i' = \frac{\theta(32 + 6\theta/\tau)}{(13 + 8\theta/\tau)}$	$\tau_d' = \frac{4\theta}{(11 + 2\theta/\tau)}$

Figure 14: Cohen-Coon (CC) Tuning Rules

<u>Controller Type:</u>	<u>Proportional Gain:</u>	<u>Integral Time:</u>	<u>Derivative Time:</u>
P-Only	$K_c = \frac{1}{K} \left(\frac{\tau}{\theta} \right)$		
PI-Control	$K_c = \frac{0.9}{K} \left(\frac{\tau}{\theta} \right)$	$\tau_I = 3.33\theta$	
PID (actual)	$K'_c = \frac{1.2}{K} \left(\frac{\tau}{\theta} \right)$	$\tau'_I = 2\theta$	$\tau'_D = \theta/2$

Figure 15: Ziegler-Nichols (ZN) Tuning Rules